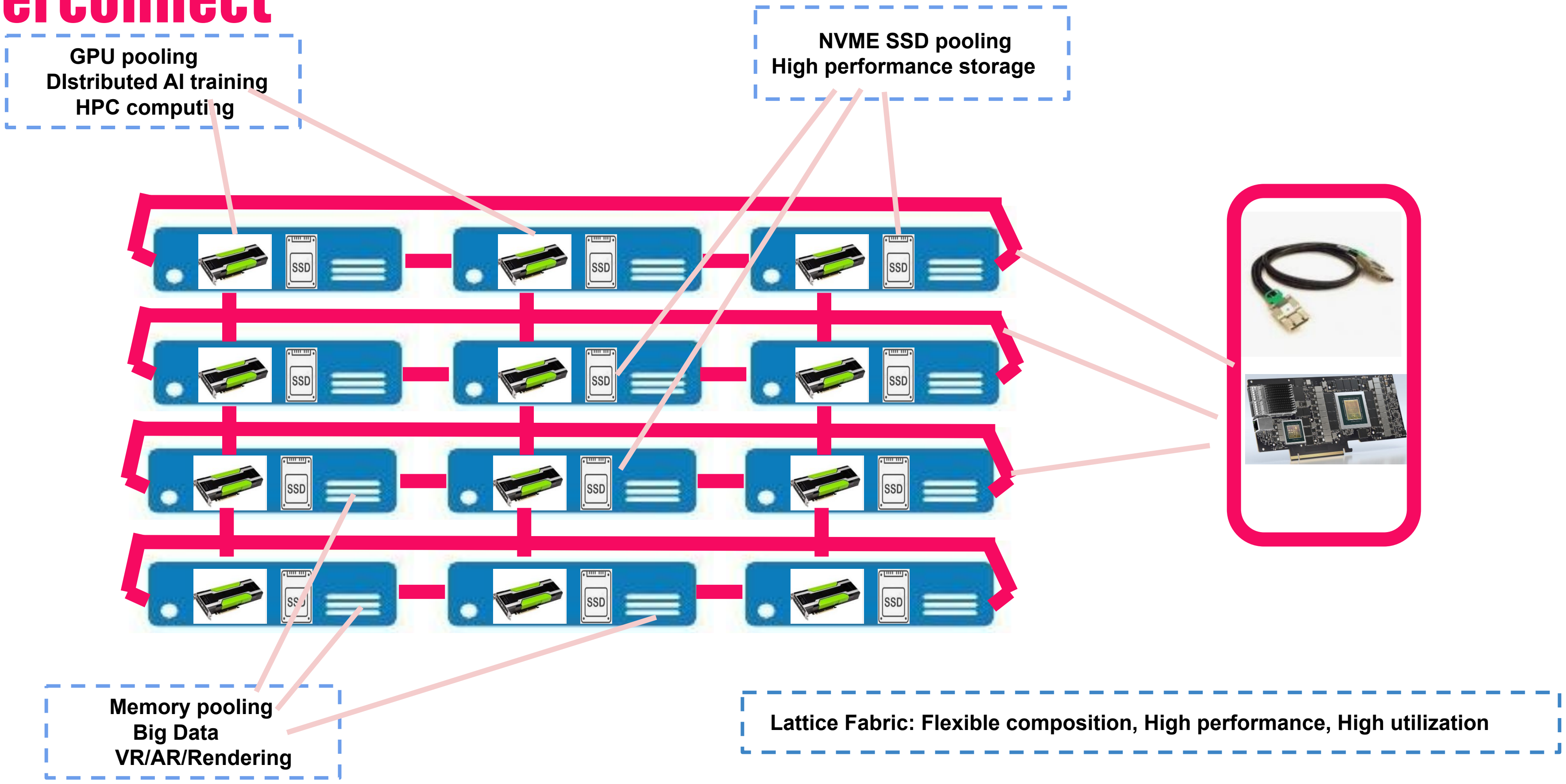
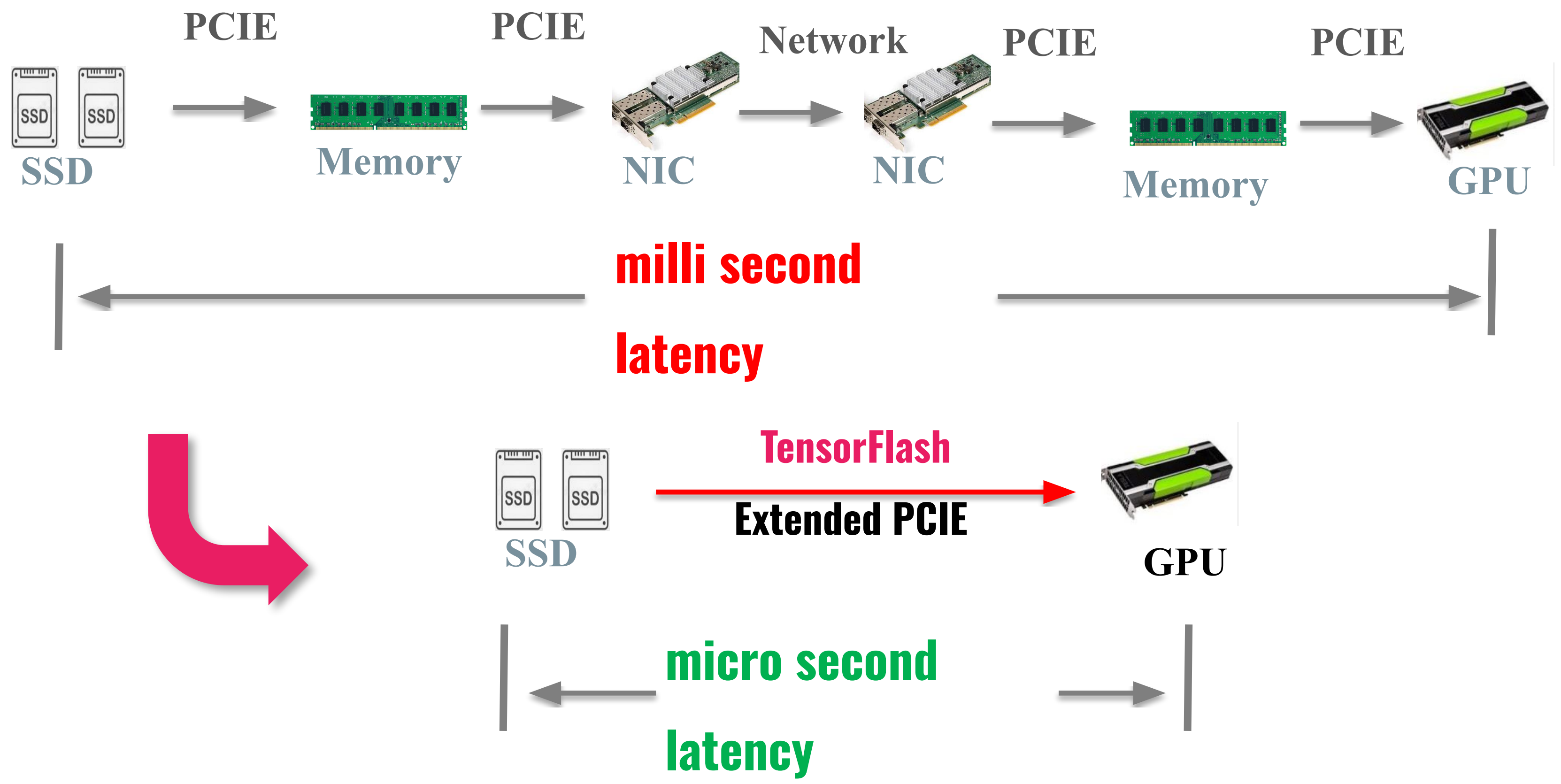
TensorFlash

Democratizing AI Infrastructure

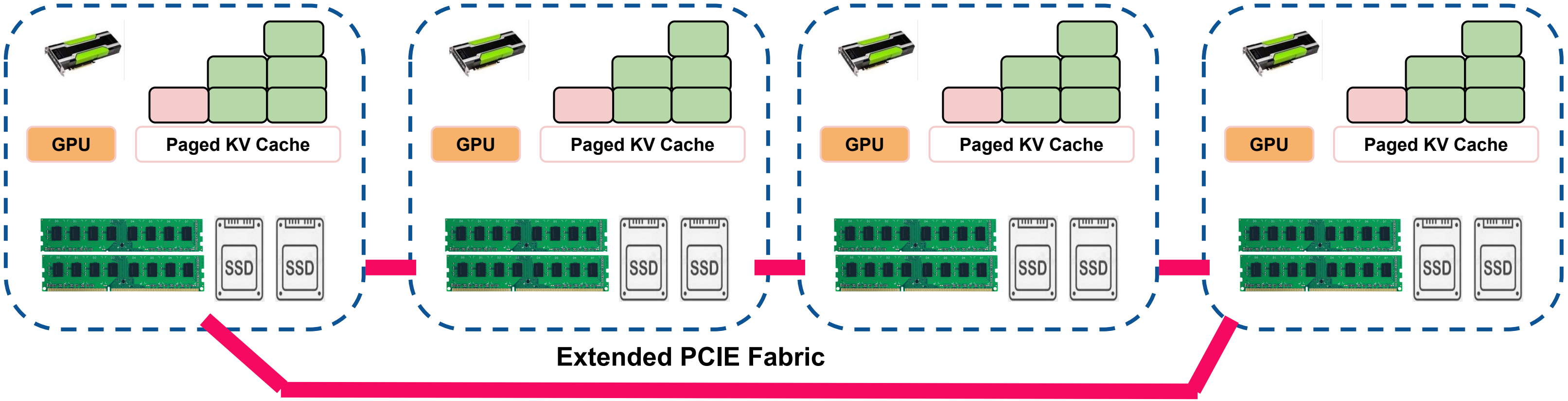
TensorFlash Innovation: Extending PCIE for Server Interconnect



Drastic Reduction in GPU-SSD Data Latency

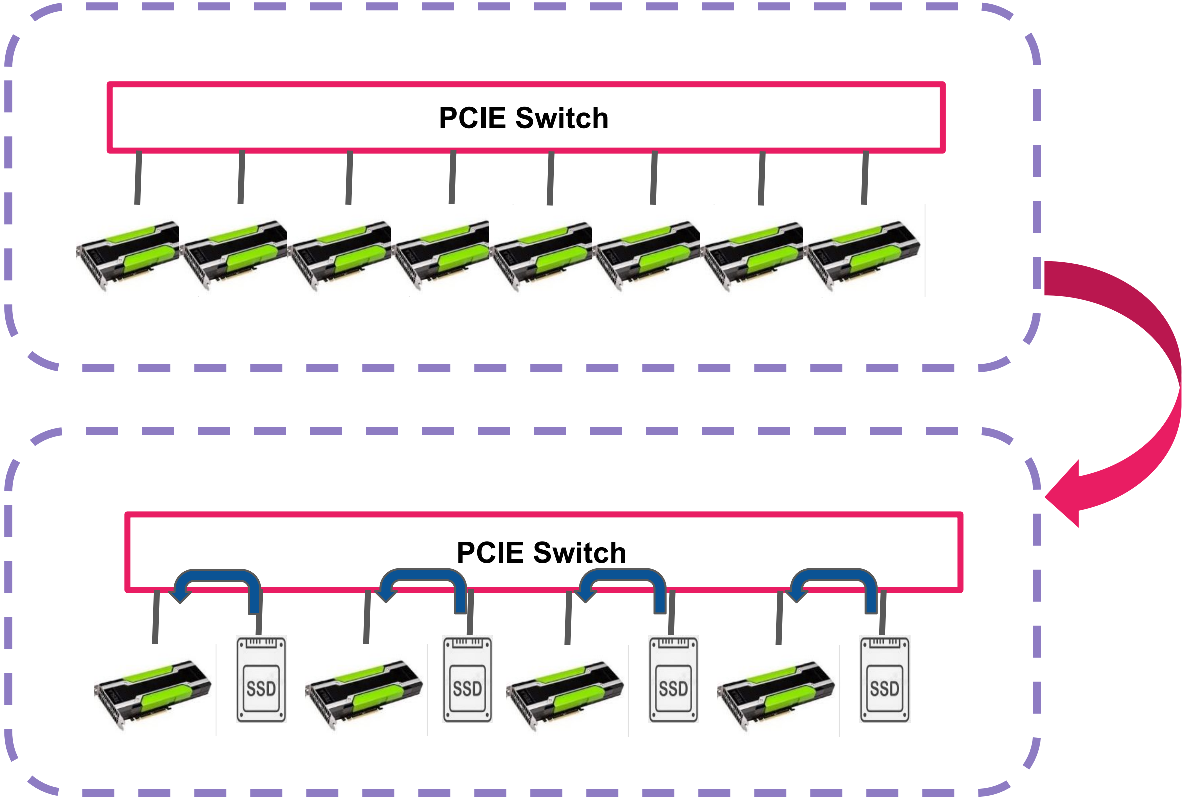


Tightly Coupled GPU/SSD KV Cache for LLM



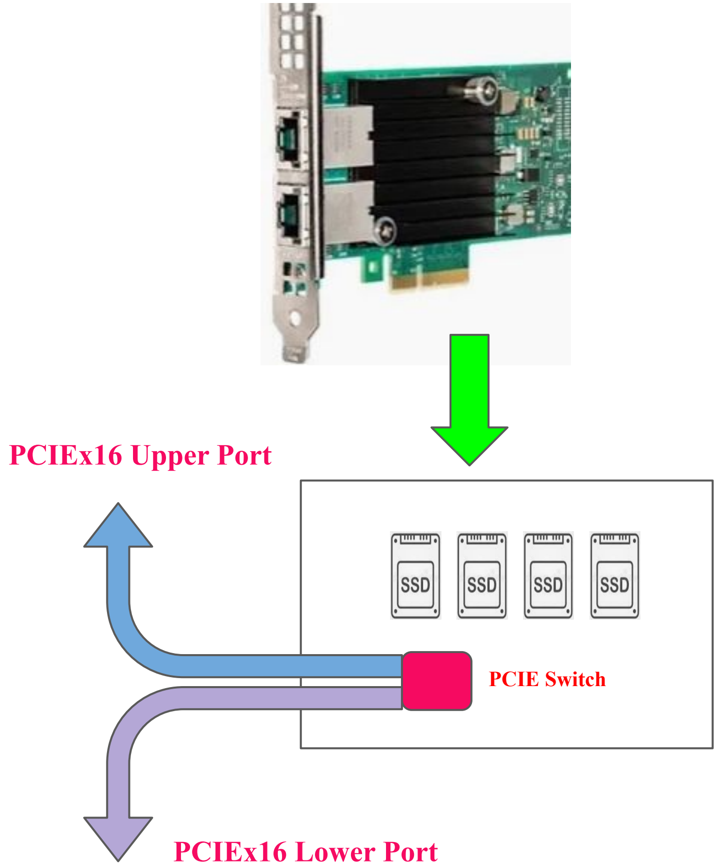
- Extended PCIE fabric provides pooling for memory and SSD
- Data movement all in hardware, minimal driver/FW involvement
- Drastic improvement in bandwidth and latency compared with traditional RDMA

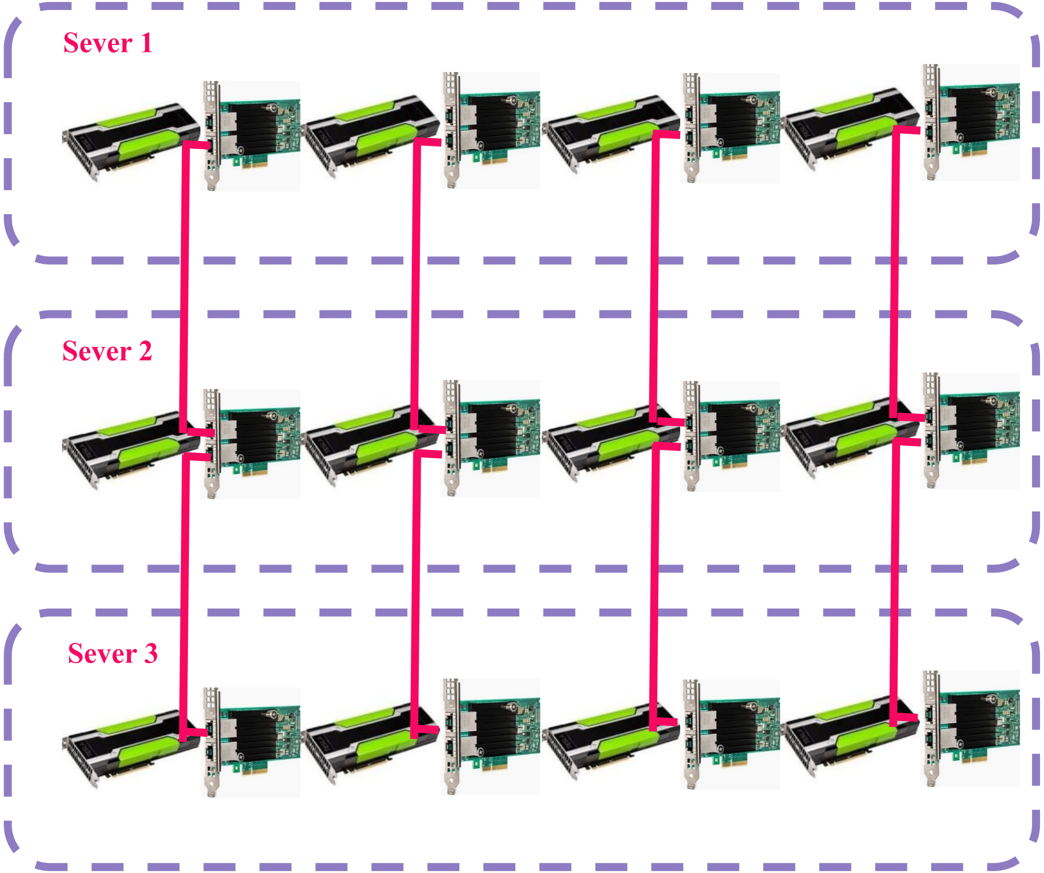
The Revolutionary TFNative™ Technology



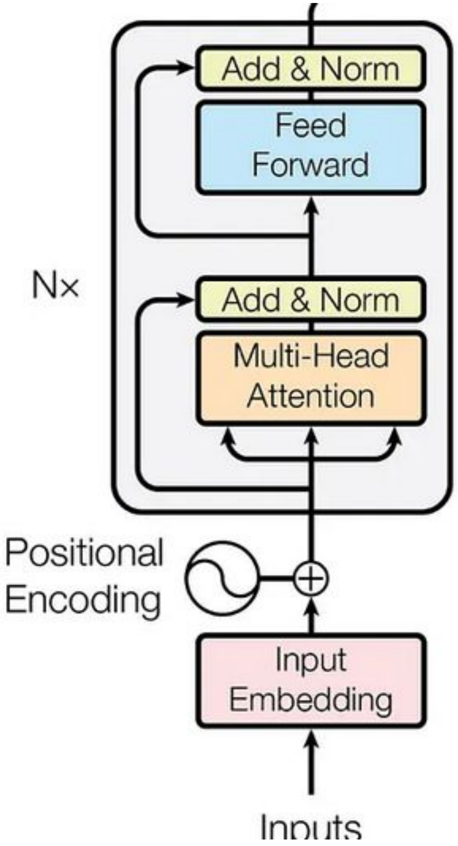
- Each GPU paired with SSD
- GPUs have dedicated computing scratch pad
- SSD hold all model parameters, gradients, optimizer states
- GPU exchange data through neighboring SSD

TensorFlash storage interconnect





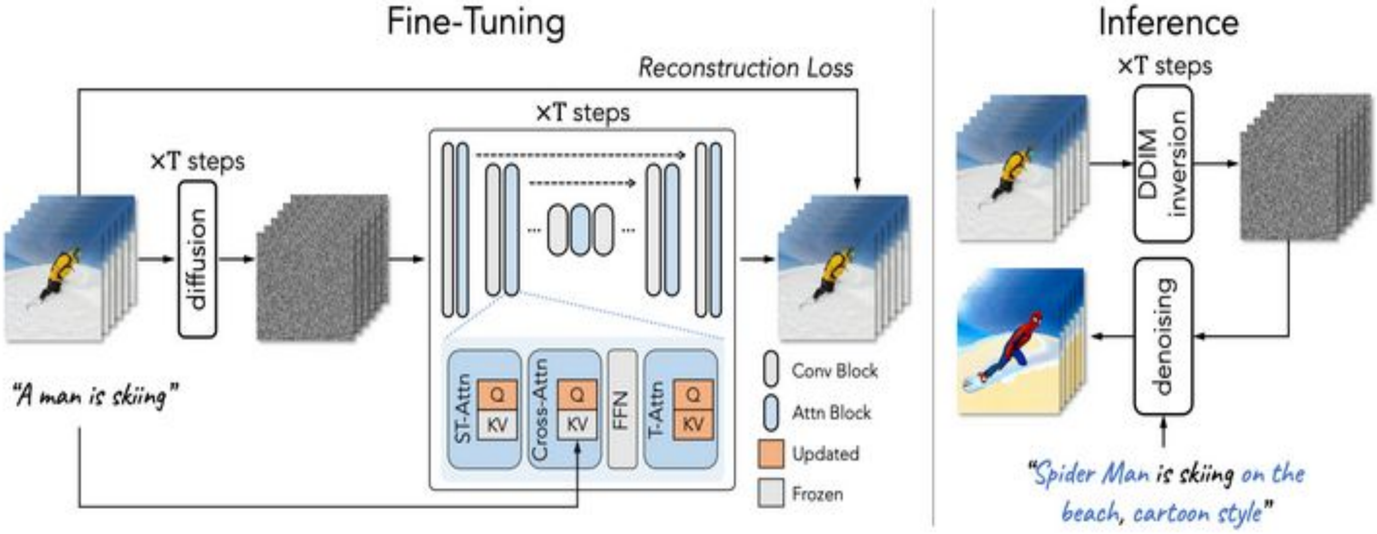
Comparable Performance with Far Fewer GPUs



Case Study of Streaming Computing during LLM Inference

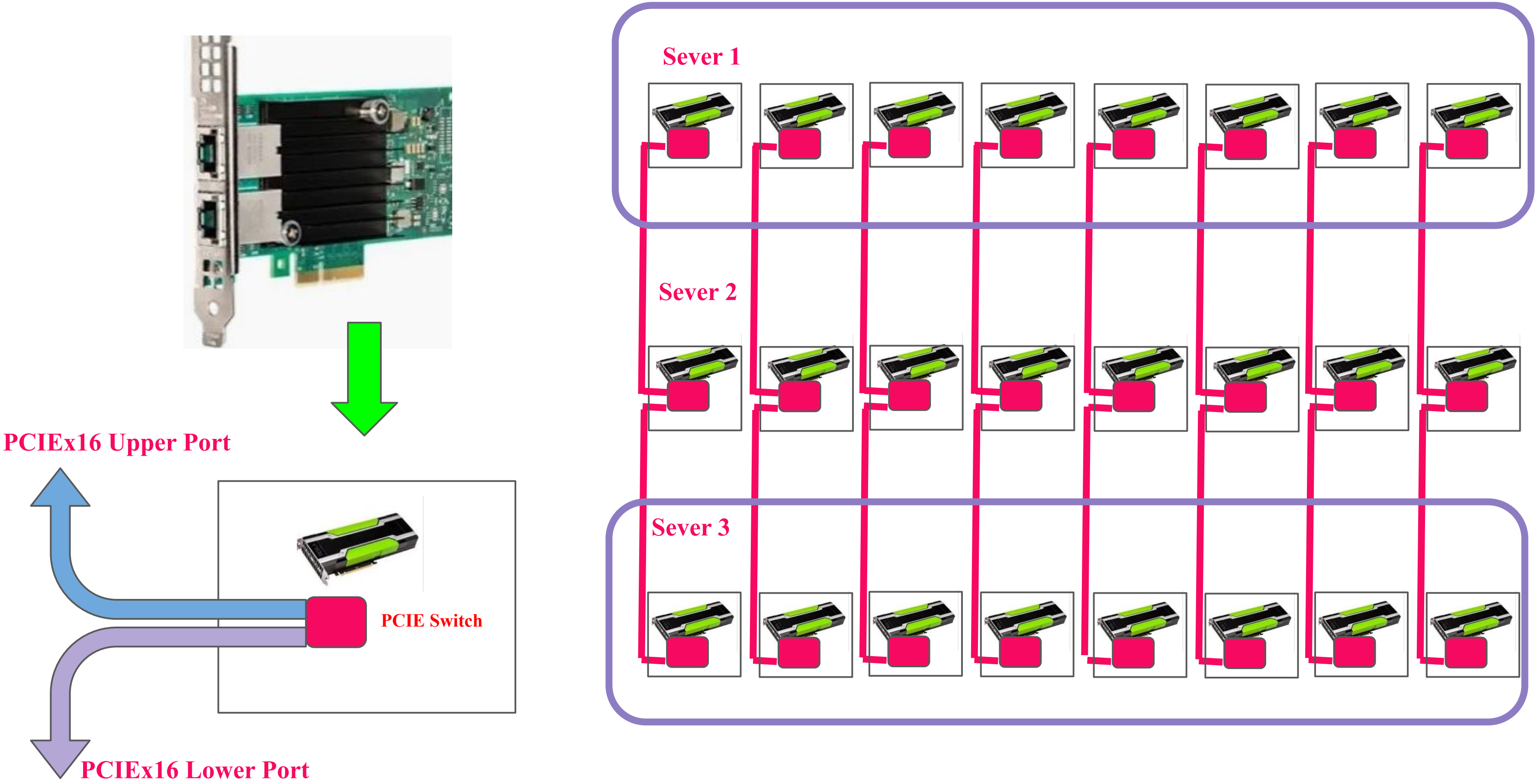
- GPT-3 175B, 96 layer
- Each layer ~1.8B, 3.6G Bytes
- Gen6x16 SSD bandwidth ~ 100GB/s
- Model loading time for one layer: 3.6/100 ~ 35ms
- Computing: 100K x 1.8B / 1000 T ~ 180ms

Diffusion Transformer (DiT) - Video Generation

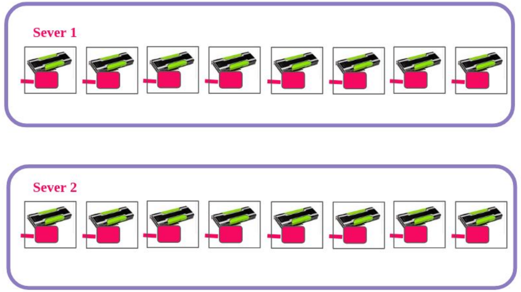


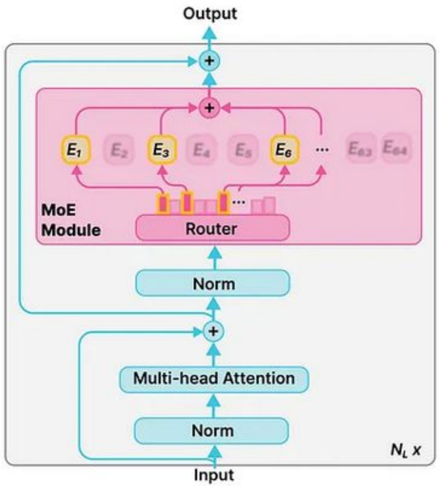
- Both Inverse and Denoise processes have long computing to overlap with IO loading
- High definition video generation requires large HBM in current approach
- TensorFlash provides high quality video generation solution with cost efficient GPU clusters

TensorFlash GPU Interconnect



DeepSeek 2 server 16 Card Inference

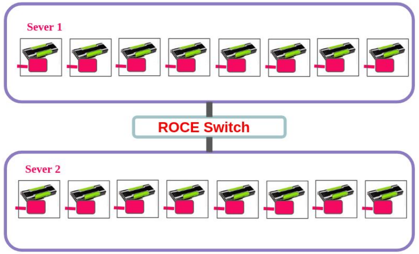


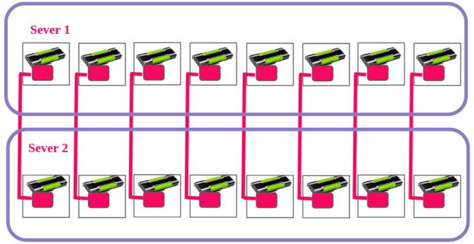


Case Study of DeepSeek V3 671B, 37B active

- DeepSeek V3 671B, 37B active, 61 layer
- Hidden Dimension(D_model) 7168
- Top_K 8
- 1K input token
- Computing time ~ 2 x 1K x 37B/150T/16 ~ 30ms

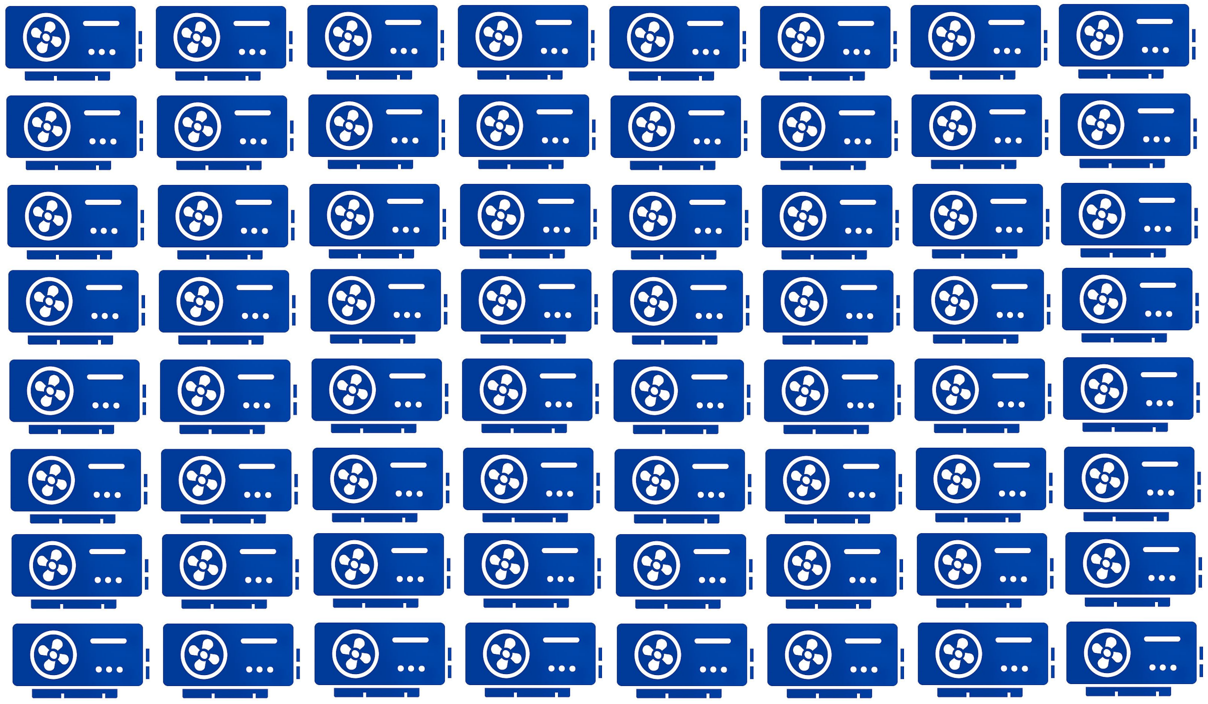
DeepSeek 2 Server 16 card comparison





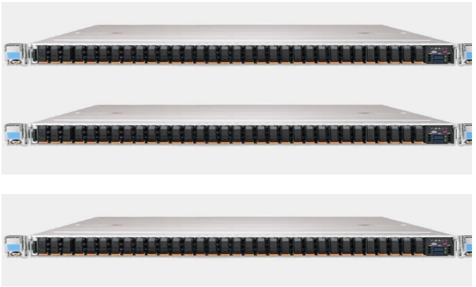
Bandwidth	400G	3.2T
All2All	2x1024x7168x61x8/50G ~ 140ms	2x1024x7168x61x8/400G ~ 17.5ms
Prefill Time	140 + 30 = 170 ms	17.5 + 30 = 47.5 ms
Token/s/GPU (input)	1k/16/0.17 ~ 368	1k/16/0.048 ~ 1330

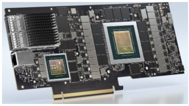
Large-scale Training with Low-cost GPU or Alternative



- Nvidia 5090 ~ $2K
- 256 5090 ~$500K
- Aggregated ~ 50 Pflops
- Train any of the largest LLM
- Multi-Agent Reinforcement Learning

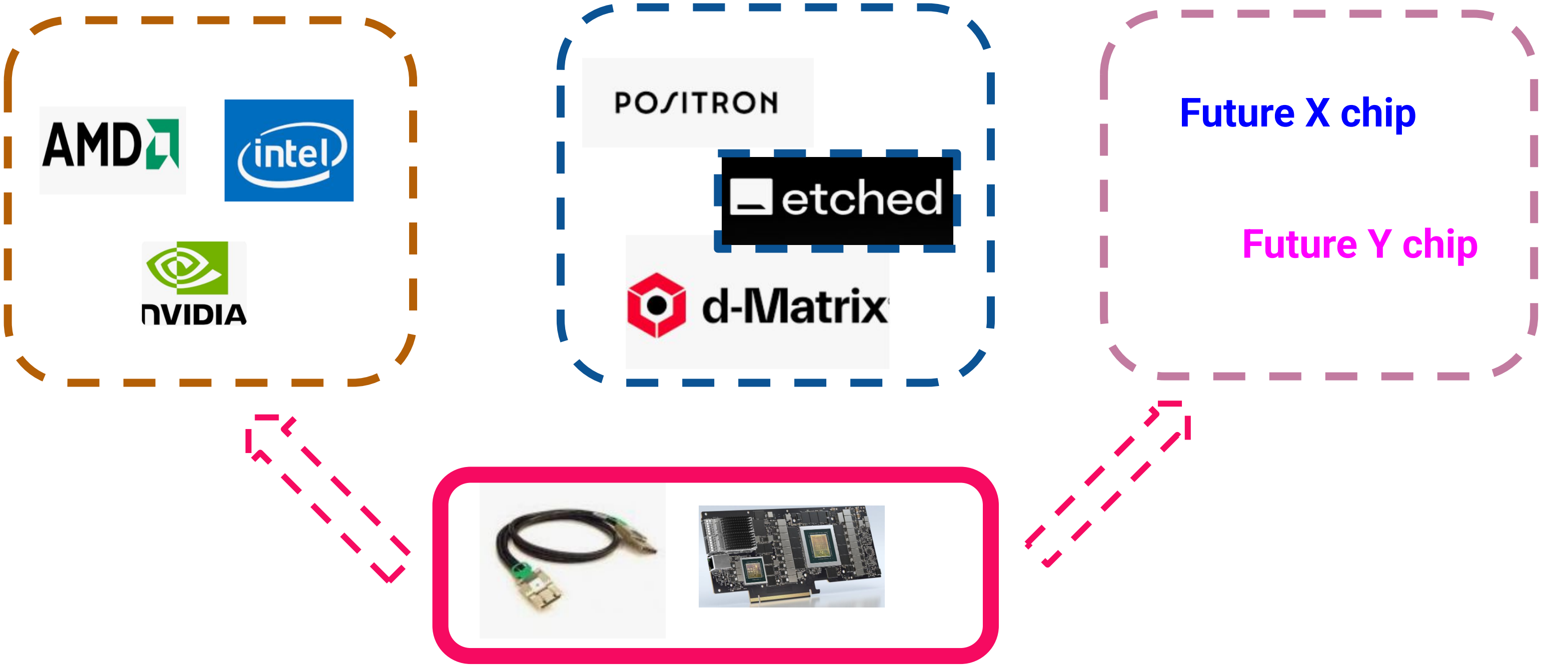
Seamless Integration with All Major Platforms





- Extended PCIE fabric turns any server platform into high performance cluster
- All Flash Array built from standard off-the-shelf components
- Converged and native SSD performance across cluster

GPU/LPU EcoSystem

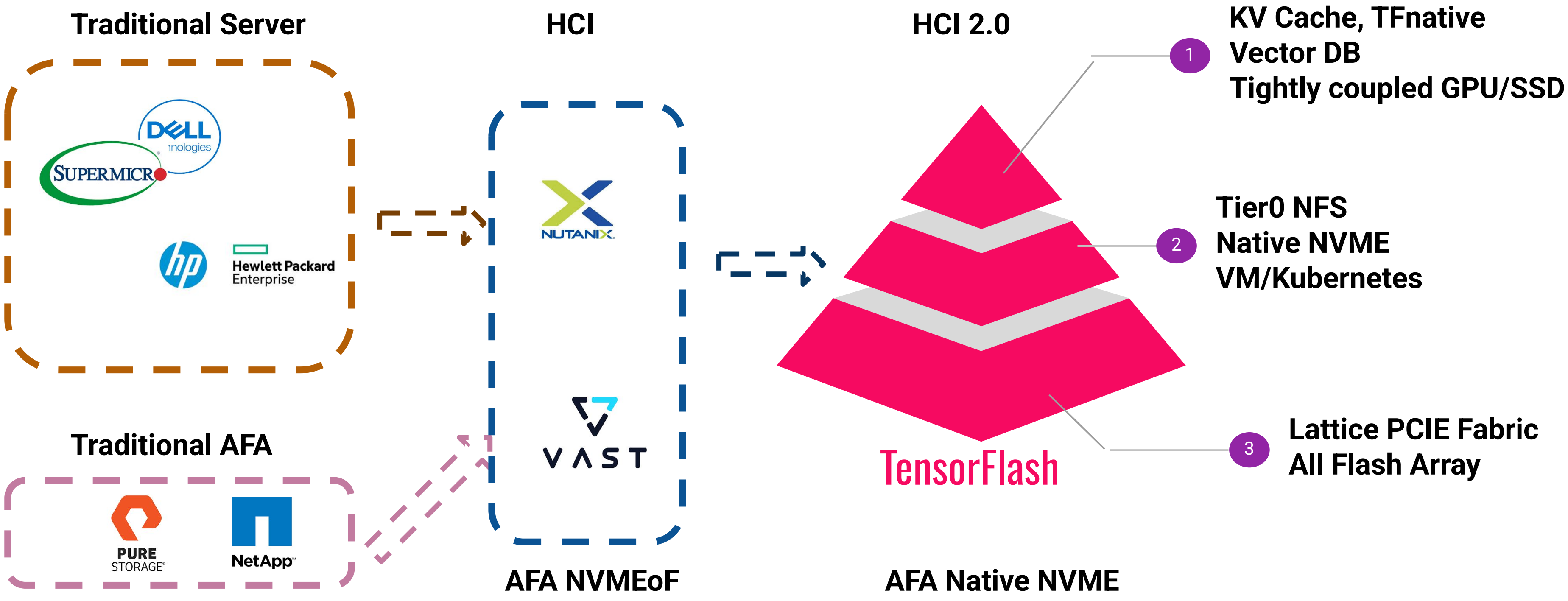


- PCIE is universal interface with all modern GPU/LPU
- TensorFlash naturally provide high bandwidth interconnect fabric

TensorFlash Computing/Price Ratio vs Competitors

GPU	AMD 9700 Intel Arc B79	Nvidia H200
Computing Power	190 Tflops	990 Tflops
Price	$1300	$30,000
	1x Tflops 1x Price	5x Tflops 20x Price

Evolution of Computing Infrastructure



Founding Team

- **Wei Zhou**, worked in Storage industry for over 20 years, have held senior management position at Marvell and SK hynix, leading teams in storage HW/FW/system engineering. Master degree of Applied Physics from Stanford

- **David Chen**, former Vice President and Head of SSD Business at Marvell, founding President of InnoGrit, founding Partner of El Camino Capital. PhD in Electrical Engineering from UC Berkeley

- **K. Wong**, worked in various fields in storage, have broad experience in SSD FW engineering and linux kernel engineering, have held principle engineer positions at Marvell and Sk Hynix

- **X. Hu**, 20-plus years' experience in storage industry, strong kernel and OS engineering expertise, have held various engineering and management positions at Marvell and startups